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Nozzle and needle during high viscosity adhesive jetting based on piezoelectric jet dispensing

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Abstract

A piezoelectric impinging jet valve is used as a study object to investigate the effect of the ball needle in the existing impinging jet and nozzle structure of the valve on the performance of the jet. First, FLUENT software is used under different ball needle and nozzle structural parameters to simulate the pressure distribution that the ball needle and nozzle in the pressure cavity form when the ball needle hits the nozzle, by arranging the structure model of the ball needle and impact valve nozzle. The piezoelectric impact injection valve and the experiment test system are then designed. Test results show that the ball needle and nozzle structural parameters are closely related to the injection performance of the impact valve. Under certain conditions, a greater needle radius corresponds to a smaller nozzle aperture and taper. Moreover, high-viscosity liquid jetting is easily achieved. By using a ball needle with a radius of 1.5 mm, a taper angle of 60°, and a nozzle diameter of 0.1 mm, we can realize the industrial viscosity of 58 000 cps in glue spray, and the injection plastic fluid volume is 0.62 μl .

Keywords: needle, nozzle, high-viscosity injection, piezoelectric

(Some figures may appear in colour only in the online journal)

1. Introduction

With the rapid development of electronic technology, electronic products are constantly being developed for intelligence, miniaturization, and diversification. The dispensing allocation method has been successfully applied to electronic packaging to satisfy the requirements of the electronic market, and dispensing technology has become the key technology of electronic packaging [1, 2]. Given the high velocity, precision dispensing, absence of z-axis movement, low noise, and other characteristics of the piezoelectric jet-dispensing valve, it is gradually being accepted as a replacement for the traditional contact-jetting dispenser [3, 4]. The piezoelectric jet-dispensing valve uses a mechanical amplification mechanism for piezoelectric stack-displacement amplification to drive the ball needle into hitting the nozzle movement to achieve jetting. Nguyen [5] proposed a piezoelectric jet-dispensing valve

based on a flexible-zoom lever. After the piezoelectric stack is amplified by the lever, the needle maximum output displacement is 0.4 mm, and the maximum injection frequency is 700 Hz. Jetting on Loctite FP4451 can be achieved when the valve is heated to 70 °C. Nguyen [6–8] proposed a piezoelectric jet-dispensing valve that is based on the principle of hydraulic amplification. The maximum velocity of the striker is 4.53 m s⁻¹ and its maximum injection frequency is 585 Hz. Glue-viscosity jetting of 10 000 cps can be achieved when the valve is heated to 80 °C. Wang [9, 10] proposed a piezoelectric jet-dispensing valve based on double-stacked amplification. After the piezoelectric stack is amplified, the maximum output displacement of the ball needle is 0.323 mm, and the maximum injection frequency is 65 Hz. The ball needle can be realized on the glue injection with a viscosity of 1000 cps and a spray-glue dot diameter of 1.07 mm. Ham [11] proposed a piezoelectric injector using a

diamond hinge to amplify the displacement of the piezoelectric stack. Analyzing the power performance of enlarged structure obtains a maximum amplification displacement of 0.330 mm; Guiling *et al* [12–14] analyze the flow of glue through the nozzle aperture when jetting high-viscosity glue. The effect of temperature, ball needle stroke, and plastic pressure on injection performance was analyzed, and showed that the larger the liquid viscosity, and the higher the temperature needed for heating, the larger the ball needle stroke. Zhang [15] conducted a theoretical analysis of the chamber, which is composed of the nozzle radius in the percussive injection valve, the ball needle, and the inner wall of the cavity. Rules on the influence of length and radius of the nozzle hole, as well as the chamber diameter and length on jet effect, are then drawn. However, they did not analyze the chamber that formed between the ball needle and nozzle orifice. Moreover, this study was limited to theoretical analysis, and the actual injection effect is not presented. These researchers did not use high-viscosity spray glue. With high dispensing process requirements, the required injection valve has high-viscosity glue. Thus, a high-viscosity injection valve is needed.

Previous studies were mainly focused on amplifying the piezoelectric stack output displacement to obtain a high frequency, spray viscosity, and ball needle output displacement. However, improving the movement of loaded needle spray glue to further improve the viscosity is difficult. The present study improves the viscosity through changing the nozzle spray glue striker structure. Detailed theoretical analysis and experimental testing of the shape and size of the ball needle and nozzle have never been presented. Therefore, this study focuses mainly on the simulation and experimental analysis of the ball needle in the piezoelectric stack impact injection valve, as well as the shape and size of the nozzle, to obtain the injection effect. The volume and velocity of liquid injection can be accurately controlled through changing the radius of the ball needle and nozzle configuration parameters, and varying the above parameters can achieve high-viscosity glue injection.

2. Principle of jetting

The impinging jetting-dispensing valve can be divided into three steps.

- i. In the normal shutdown state, the impinging jetting-dispensing valve is normally a closed valve. The ball needle and the nozzle are in a close-fit state with the role of sealing, which can effectively prevent the glue in the tract from flowing out through the nozzle orifice, as shown in figure 1(a).
- ii. The ball needle separates from the nozzle when raised using the driving force, and the glue in the channel is transported to the gap between the ball needle and nozzle under pressure (figure 1(b)).

- iii. The third step is the hitting process. When the ball needle moves down and hits the nozzle, an instantaneous high pressure can be formed between the ball needle and nozzle, and this high pressure forces the glue to flow fast toward the nozzle orifice. Meanwhile, the ball needle cuts off the glue in the channel that flows to the nozzle when the nozzle is hit. Thus the instantaneous pressure generated will force the glue at the nozzle orifice to remove the fluid force at the spray tip, and then the injection is realized (figure 1(c)).

The selection of the ball needle and nozzle is also important because the objects should be in a close-fit state in the normal shutdown state, which can effectively prevent glue leakage with the function of sealing. Three kinds of coordination mode are used in the ball needle and nozzle: conical surface to conical surface, spherical surface to conical surface, and spherical surface to spherical surface. The structure of a spherical ball needle and a conical nozzle are chosen in this study.

3. Theoretical analysis of jetting

3.1. Analysis of jetting velocity

The ball needle and the nozzle press on the glue when the ball needle hits the nozzle, and this high pressure drives the glue to overcome its adhesion and jet through the nozzle orifice. A study found that the adhesive solution accumulates in the nozzle and cannot overcome the surface tension to jet out when the jet velocity is small. Thus, the velocity of the glue solution through the nozzle orifice must be greater than a critical value v to achieve a stable injection. For the non-Newtonian fluid, the fluid layers have a relative moment, and viscous force F_μ produced at this time can be expressed as follows:

$$F_\mu = v\mu d \quad (1)$$

where μ is fluid dynamic viscosity, v is liquid velocity, and d is the nozzle diameter.

The glue at the nozzle will produce a certain velocity when the ball needle hits downward. Under the effect of inertia, the adhesive solution overflows. At this time, the glue inertia force F_i can be expressed as

$$F_i = \rho v^2 d^2 \quad (2)$$

where ρ is glue density, thus the necessary condition to eject the glue is

$$F_i > F_\mu \Rightarrow v > \frac{\mu}{\rho d}. \quad (3)$$

The above formula shows that the glue velocity must exceed a critical value v for the adhesive solution with a certain viscosity to achieve injection. The greater the glue velocity is, the larger the kinetic energy the droplet gains, the easier it is for the glue to overcome its own viscous force and spray out, and the greater the glue viscosity that is sprayed

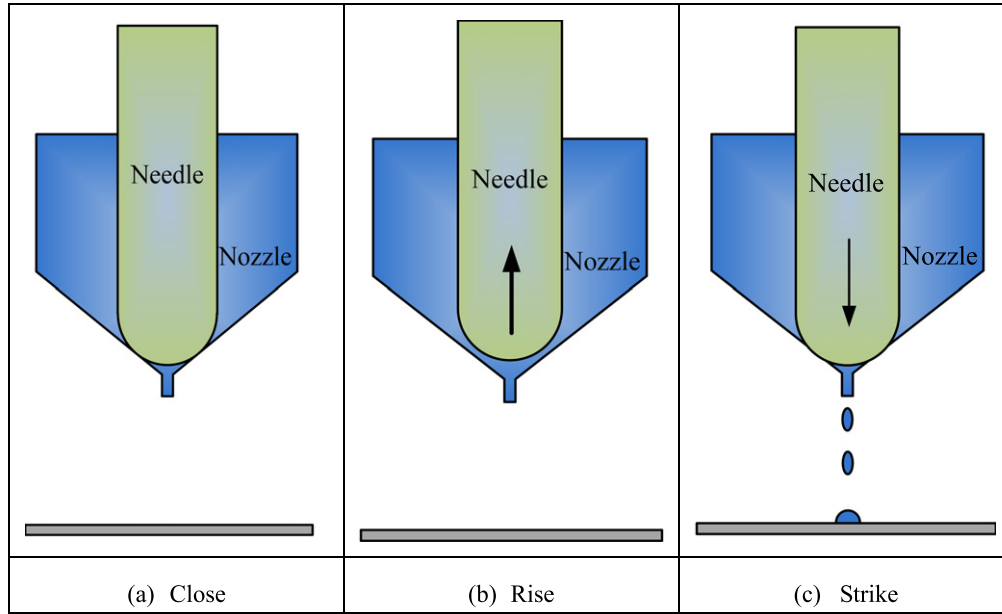


Figure 1. Process of impinging jet.

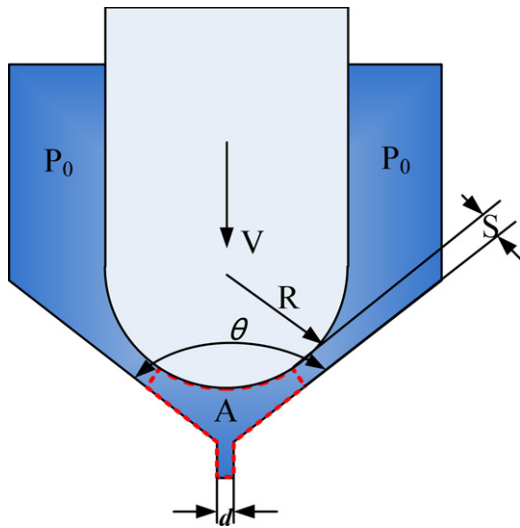


Figure 2. Simulation model of the ball needle and nozzle.

out. Therefore, improving the velocity of glue at the nozzle is the key problem for design of the jetting dispensing valve. The fluid can be ejected as fine fluid streams because of the small size of the nozzle. The fluid jetting at the nozzle is free flowing from the hole under the pressure, and the fluid jet velocity [16] is

$$V_{\text{jet}} = c_p \sqrt{\frac{2\Delta p}{\rho}} \quad (4)$$

where c_p is the flow rate coefficient for holes; Δp is the average pressure difference between the two ends of the holes.

4. Flow field of jetting chamber simulation

The simulation model of the ball needle hitting the nozzle is established using Gambit software, as shown in figure 2. The divided grid model is then imported into FLUENT, and the action grid is analyzed using the self-editing UDF function. Set the boundary conditions with the inlet pressure 0.2 MPa, and the outlet pressure 0.1 MPa. Some parameters of the adhesive are shown in table 1. The ball needle spherical radius is 1.25 mm (all ball needles are expressed in radius), the nozzle angle is 120° , and the nozzle radius is 0.2 mm. The stroke of the ball needle is 2 mm, with 0 mm s^{-1} as initial velocity to accelerate to hit the nozzle; the ball needle's velocity when hitting the nozzle is 3 mm s^{-1} ($t = 0.133 \text{ s}$). The pressure distribution inside the jet chamber at the nozzle is shown in figure 3. Figure 4 is the pressure graph of the glue inside the jet chamber when the ball needle hits the nozzle.

Figure 4 shows that the process of the needle moving down from the highest point when the needle and nozzle are far apart squeezes the glue, which causes most of the glue from the medial wall of the needle and inclined side wall of the nozzle to flow upward. At this point, the pressure at the needle and nozzle is mainly supply pressure accompanied by a small increase. Under the influence of surface tension and viscous forces, the glue cannot pass through the gap and flow upward when the gap between the needle and nozzle sidewall is small enough, which suddenly increases the glue pressure inside the jet chamber. The curve in figure 5 shows that the glue pressure inside the jet chamber increases significantly. Meanwhile, the injection cavity pressure increases instantaneously and then immediately decreases because of the pressure that acts on the glue, that is partly converted into liquid kinetic energy. The glue injection cavity fluid pressure then steadily grows as the gap between the needle and nozzle becomes smaller. When the needle hits the nozzle such that

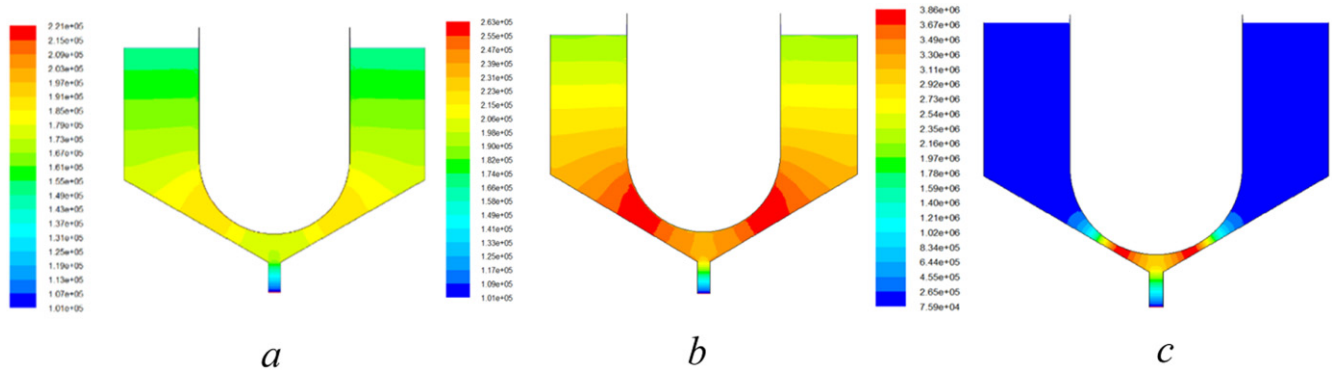


Figure 3. Pressure distribution diagram of the area between the ball needle and nozzle. (a) The moment before the ball needle hits the nozzle. (b) The moment that the ball needle hits the nozzle. (c) The moment after the ball needle hits the nozzle.

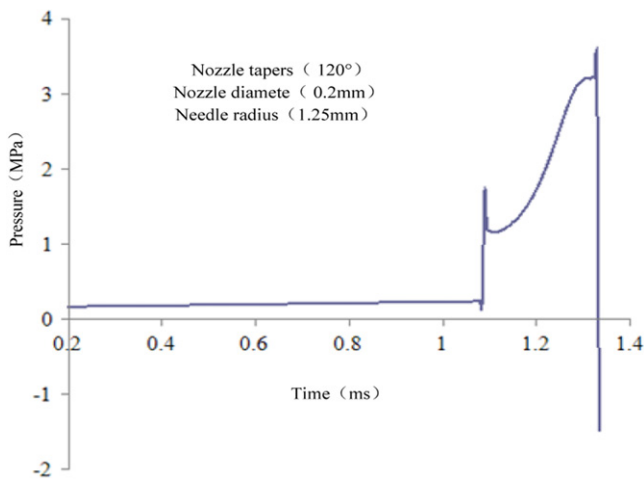


Figure 4. Pressure curve of the area between the ball needle and nozzle.

they fit completely and cuts off the other glue flow into the injection chamber, the glue flow rate of the injection orifice suddenly becomes zero, and the injection pressure in the chamber becomes negative. Therefore, the glue outflow also sprays onto the substrate at the same velocity.

With the above simulation boundary conditions, the pressure variation curve of the glue in the injection chamber between the needle and nozzle of the jetting chamber can be obtained at different radii of the needle, which is shown in figure 5.

The simulation results of figure 5 show that the pressure between the needle and the injection chamber gradually increases with increasing needle radius. The use of a large-radius needle can generate great instantaneous pressure in the injection chamber A to achieve a high-viscosity glue injection, because the higher the viscosity of the glue is, the larger the viscous forces between the droplets are. Thus, sufficient pressure must be obtained to cut off the viscous force between the droplets. Therefore, this simulation concludes the following: high-viscosity adhesive liquid injection can be achieved through increasing the radius of the needle. Meanwhile, great instantaneous pressure can be generated using a needle with a large radius. Thus, the net amount of glue that flows out of the injection chamber is great. Within a certain

range, a jet that dispenses a large-volume, high-flow, and high-viscosity fluid can be achieved using a large needle.

This study found that the nozzle taper is also one of the key factors that affect injection. Therefore, in-depth research is necessary. Without changing the other structure parameters, the simulation is conducted through changing the nozzle angle, and the curve can be obtained as shown in figure 6.

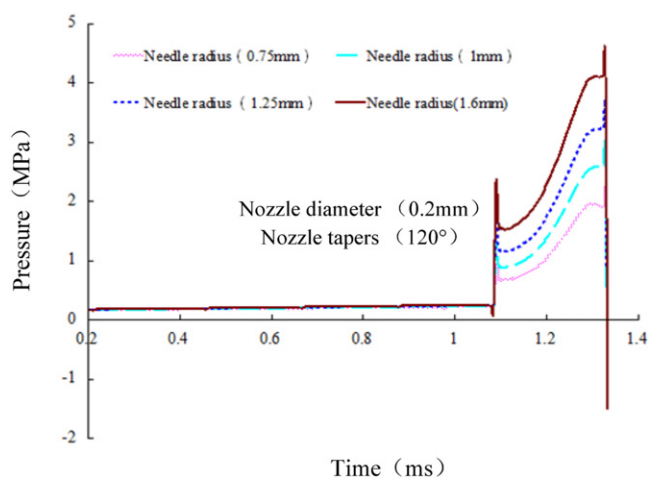
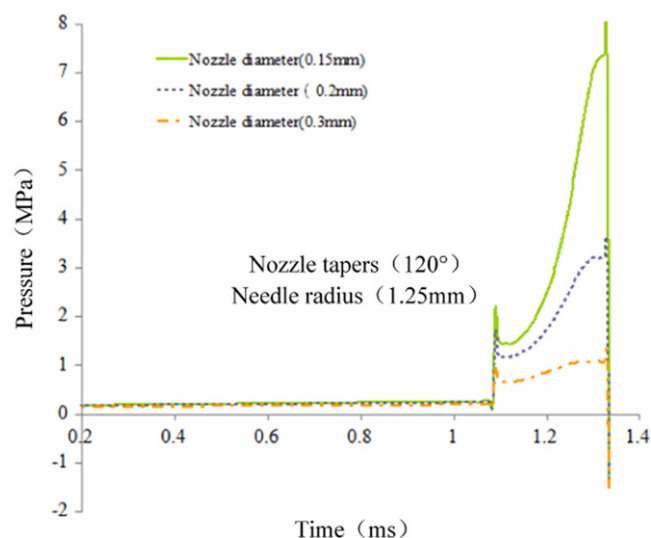
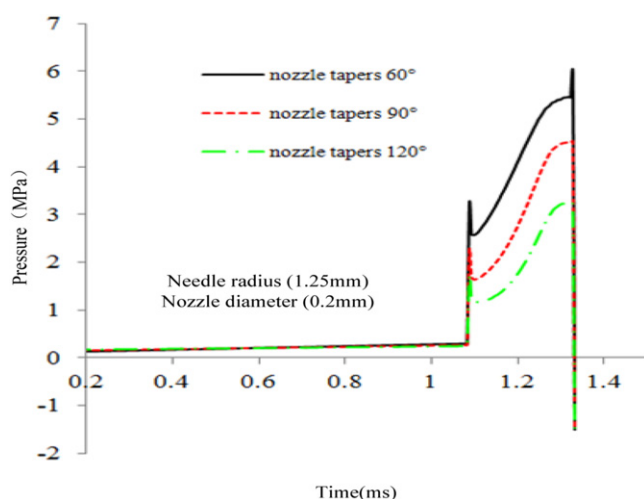
As shown by the simulation in figure 6, a smaller nozzle taper θ corresponds to a higher instantaneous pressure generated when the needle hits the nozzle. This result is attributed to the fact that a smaller nozzle taper leads to a smaller gap S generated when the needle hits the nozzle. The total flow and the nozzle orifice radius d are constant; thus, a high flow velocity in the injection cavity easily produces high pressure. The smaller the nozzle taper θ , the higher the instantaneous pressure generated, and the glue obtains higher instantaneous kinetic energy, which allows for the liquid viscous force between the glue droplets to be cut off easily. Therefore, a high-flow and large-volume injection can be easily achieved. Meanwhile, high-viscosity glue injection can be achieved through changing the nozzle taper.

Meanwhile, the diameter of the nozzle orifice is one of the key factors affecting the glue injection cavity pressure. Therefore, the glue pressure distribution in the injection cavity can be obtained through analyzing different nozzle orifice diameters with the premise of not changing other structural parameters and boundary conditions, as shown in figure 7.

Figure 7 shows that the smaller the nozzle hole diameter is, the higher the instantaneous pressure generated in the injection cavity is. This phenomenon occurs because the glue encapsulated by the needle and nozzle flows out through the nozzle orifice and the gap between needle and nozzle when the needle moves down and hits the nozzle. However, the gap between needle and nozzle becomes smaller as the needle gradually moves downward, and most of the glue flows out from the nozzle orifice. Thus (within a certain range), the smaller the nozzle orifice is, the higher the ejection velocity generated at the nozzle opening is, and the higher the instantaneous pressure is in injection chamber A, which is between needle and nozzle. Therefore, the volume of spray glue can also be adjusted further to control the dispenser

Table 1. Some parameters of the adhesive

Density $\rho(\text{kg m}^{-3})$	Viscosity $\mu(\text{cps})$	Surface tension coefficient $\sigma(\text{mN m}^{-1})$	Yield stress $\tau(\text{Pa})$
1300	1000	20	150

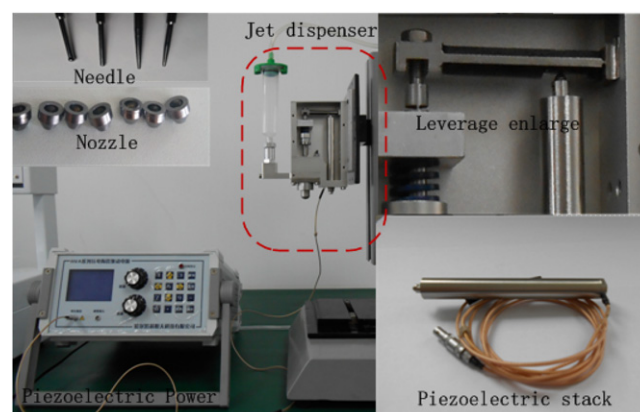
**Figure 5.** Pressure distribution for different needle radii.**Figure 7.** Pressure distribution for different nozzle hole diameters.**Figure 6.** Pressure distribution of different nozzle tapers.

effects through changing the parameters of the nozzle orifice diameter; high-viscosity adhesive solution injection can also be achieved.

5. Experiments and discussion

5.1. The establishment of the experimental system

In this study, the experimental system is composed of the jet-dispensing valve device driven with piezoelectricity, piezoelectric stack drive power, a precision electronic scale, and other components. The piezoelectric jetting dispensing valve device (figure 8) mainly uses a flexible lever to amplify the displacement of the piezoelectric stack to drive the needle to hit the nozzle. The quality of the ejected dot can be measured

**Figure 8.** Experiment test system of the hydraulic and piezoelectric-driven jet-dispensing valve.

and its volume can be finally calculated using a precision electronic scale.

From the above analysis, we can conclude that the same injection radius, with the premise that jetting is assured and the other conditions are the same, results in the following phenomenon: the faster the flow rate through the nozzle orifice is, the greater the volume of the injecting glue in an injection cycle is. Meanwhile, the glue flow rate needed at the nozzle orifice is great to achieve high-viscosity glue injection. Therefore, we can also improve the injection speed and increase the instantaneous pressure to achieve high-viscosity injection. Thus, the key issue is finding a method that will improve the speed and increase the instantaneous pressure.

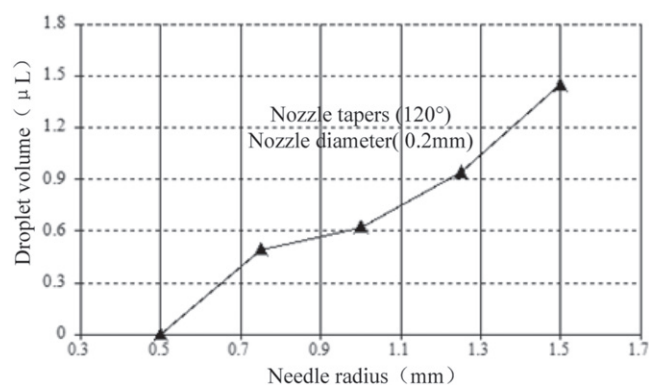


Figure 9. Relationship between needle radius and spray adhesive dot volume.

5.2. Impact of the needle radius on the injection

Industrial glycerin is used in the experiment as the injection target, with a viscosity of 1200 cps, a density of 1260 kg m^{-3} , a piezoelectric power output square wave signal to the piezoelectric stack (high level 100 V, 5 ms; low level 5 ms, 0 V). The glue supply pressure is regulated to 0.1 MPa, the nozzle taper to 120° , and the nozzle hole diameter to 0.2 mm. Different radii of ball needles are used for the experiment and the experimental graph can be gained, as shown in figure 9.

The experimental data in figure 9 show that the nozzle outlet experiences the ‘hanging plastic’ phenomenon when using a ball needle that has a radius of 0.5 mm, and the glue cannot jet out. This phenomenon occurs mainly because the instantaneous pressure generated in the injection chamber is small when the needle (a sphere with a radius of 0.5 mm) hits the nozzle, which leads to a flow rate of glue at the nozzle orifice which is less than the critical jet ejection velocity and is unable to overcome its viscous force. Thus, the single glue droplets cannot be cut off, and this results in the hanging plastic phenomenon. When the ball needle radius is more than 1.5 mm, the piezoelectric injection valve can achieve the glue injection effectively. In addition, the glue dot injection radius increases as the ball needle radius increases. This is because the traffic generated in the injection chamber A becomes larger as the needle moves downwards when the needle radius increases, and the nozzle radius d and the gap S remain the same. Thus, the flow rate at the nozzle orifice and the instantaneous pressure generated in the injection chamber A both become larger. The glue ejection volume also becomes larger. Therefore, the larger the needle radius is, the greater the instantaneous pressure generated in the injection chamber A is when the needle hits the nozzle, the higher the flow rate through the nozzle orifice is, and achieving high-viscosity glue injection is easy.

5.3. Effects of nozzle angle on the performance of the injection

The same industrial glycerin with a viscosity of 1200 cps is used in the experiment as the jet object, with piezoelectric power output square wave signals to the piezoelectric stack (high level 100 V, 5 ms; low level 5 ms, 0 V). The glue supply pressure is regulated to 0.1 MPa, the nozzle hole diameter is

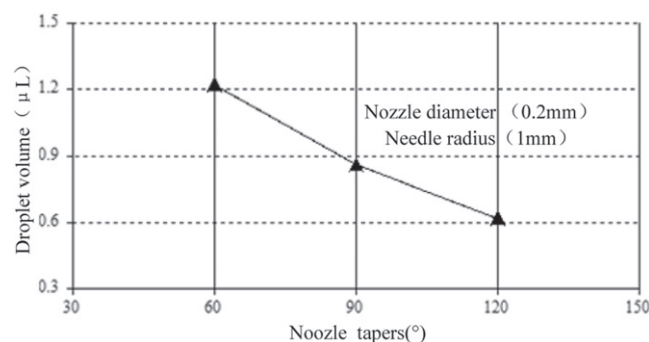


Figure 10. Relationship between nozzle tapers and spray adhesive dot volume.

0.2 mm, and the radius of the needle is 1.0 mm. Different nozzle tapers are used in the injection experiment, and the experimental graph can be obtained, as shown in figure 10.

The curves of figure 10 show that the nozzle taper and the jet glue volume are inversely proportional. The smaller the taper nozzle is, the greater the volume of ejected glue is, and the greater the jet velocity ejection generated at the jet opening is. This phenomenon occurs because the smaller the taper is, the smaller the travel gap S is between the needle and the nozzle. Thus, the glue in the ejection chamber that flows back is small when passing through the gap S . As a result, most of the glue will flow out from the nozzle orifice, thereby the flow rate produced at the nozzle orifice is high: that is, the instantaneous pressure generated in the injection chamber A is also great. Finally, the glue volume ejected is then great.

In the experiment, the nozzle cone angle is 60° when the ball needle head radius is 1.0 mm. The ‘satellite drop’ phenomenon appears easily around the glue drop that has jetted down because the glue injection speed is so fast that splashing occurs when glue droplets are ejected onto the substrate as the needle hits the nozzle. Thus, controlling the jet velocity for different viscosities of glue is important. When the jet velocity is too high, producing the ‘satellite drop’ phenomenon is easy. When the jet velocity is too small, the nozzle mouth experiences the ‘hanging plastic’ phenomenon. Therefore, the injection velocity can be changed and controlled through changing the needle and the nozzle structure in the process of glue injection.

5.4. Nozzle hole diameter

The same industrial glycerin that has a viscosity of 1200 cps is used in the experiment as the jet object, with piezoelectric power output square wave signals to the piezoelectric stack (high level 100 V, 5 ms; low level 5 ms, 0 V). The glue supply pressure is regulated to 0.1 MPa, the nozzle taper is 120° , and the radius of the needle is 1 mm. Different nozzle diameters are used, and the experimental data graph can be obtained, as shown in figure 11.

Figure 11 shows that the volume of injected glue also increases as the nozzle diameter increases. The preceding analysis shows that the larger the nozzle aperture is, the smaller the instantaneous high-pressure generated in the

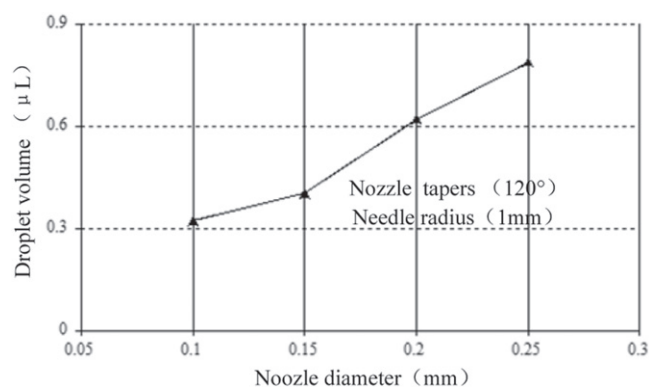


Figure 11. Relationship between nozzle diameter and spray adhesive dot volume.

injection chamber is. Thus, the flow rate of the nozzle orifice decreases. That is, the glue volume produced using the unit cross sectional area of the nozzle orifice also decreases. However, the increasing trend of the nozzle aperture area is greater than the changing trend of the glue flow rate at the nozzle hole. The glue injection volume also increases as the diameter of the nozzle increases. However, the smaller the nozzle diameter, the easier for the solid particles in the glue to become clogged. At present, the injection aperture of the general injection valve is 0.1 mm or more. Thus, a small-volume and high-velocity jet can be realized using small nozzle aperture. Meanwhile, achieving a high-viscosity glue injection is easy.

5.5. Experiments of high-viscosity glue injection

From the above experiments, we can see that the greater the needle radius is, the smaller the nozzle taper angle and the nozzle diameter are, and the higher the instantaneous pressure generated in the injection chamber is with increasing jet velocity generated. Moreover, overcoming the viscous force between the glue droplets is easy. However, the viscous force is great for high-viscosity fluid glue. To achieve high-viscosity glue injection, high jet velocity and high instantaneous pressure is needed. Thus, high-viscosity glue injection can be achieved through increasing the radius of the needle, and decreasing the nozzle angle and diameter.

Industrial glue with a viscosity of 58 000 cps is used in the experiment to perform the spray test. A needle with radius of 1.5 mm is selected, the nozzle taper is 60°, and the nozzle orifice diameter of 0.1 mm can obtain the injection effect, as shown in figure 12. Using the electronic scales we can measure the glue drops as 0.62 μL. Therefore, the piezoelectric injection valve can spray high-viscosity glue through changing the structure parameters of the nozzle and the needle.

6. Conclusions

In view of the effect of the existing impinging jet-dispensing valve needle and nozzle structure on the performance of the injection with the piezoelectric effect injection valve as the

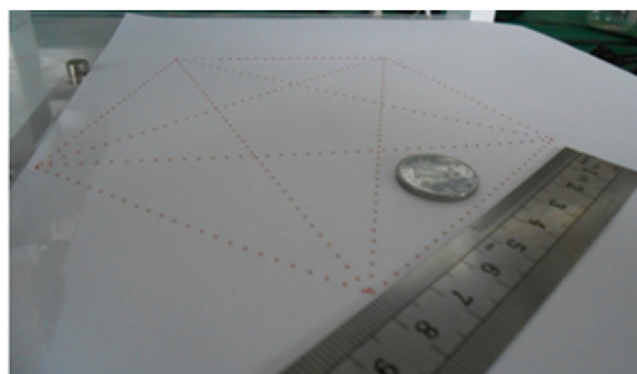


Figure 12. Glue dot plot in jetting experiments.

research object, this study explores the influence of the needle and the nozzle structure parameters on the injection performance. A needle and nozzle structure model in FLUENT software is established through changing the radius of the needle, the aperture, and the taper angle of the nozzle. We can obtain the respective pressure distribution figures formed in the injection chamber when the needle hits the nozzle. The piezoelectric impact injection valve is then designed and manufactured, a test platform is set up, and the different structures of the needle and the nozzle undergo experimental testing. This study found the following.

- (1). When increasing the needle radius, both the glue volume jet from the piezoelectric impact injection valve and the speed of glue passing through the nozzle become large, and achieving a high-viscosity adhesive liquid jet is easy.
- (2). When reducing the taper angle of the nozzle, the glue volume and the jet velocity both increase, and it is easier to achieve a high-viscosity adhesive liquid jet.
- (3). When reducing the nozzle diameter, the volume of the glue jet from the piezoelectric impact injection valve reduces. However, increasing the jet velocity is good for high-viscosity spray. When the needle radius is 1.5 mm, the nozzle taper is 60°, and the nozzle orifice with a diameter of 0.1 mm is selected. Jetting of industrial glue with a viscosity of 58 000 cps can be realized, and the spray glue volume is 0.62 μL. Of course, the focus in this study is only on the effect of the conical nozzle and ball needle structure on the jet performance. For the next step, focus should be on the effect of other nozzle shapes and ball needle structure on the jet performance.

Acknowledgement

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